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Non-radial pulsations of neutron stars with a crust

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SUMMARY
A fully relativistic treatment of non-radial pulsational modes of spherical stars with solid crusts is presented. This work complements Schumaker & Thorne's relativistic treatment of torsional modes. The strain tensor for arbitrary even-parity elastic deformations of a spherical star is derived. A Hookean law is assumed to relate the stress to the strain, and the pulsational equations and their boundary conditions are derived from the Einstein Field Equations (EFE) and the equations of motion. The EFE and equations of motion are then used to find the Lagrangian density and an action principle for the pulsational problem. This action principle may prove useful in future studies of the stability of non-radial pulsational modes for stars with a solid crust.

1 INTRODUCTION

1.1 Motivation

Within hours after its formation in the collapse of a normal star or a white dwarf, the outer layer of a neutron star solidifies to form a solid crust (*cf.* Shapiro & Teukolsky 1983 and references therein). A solid crust can support shear stresses, and this leads to two new sets of pulsational modes. The *torsional modes* are odd-parity, twisting modes of the crust: their motion is purely toroidal with no radial or poloidal component. There also appear a set of new *non-radial modes*, associated with the ability of the solid crust to propagate shear waves.

In this paper, we present a fully relativistic treatment of the non-radial modes of neutron stars with elastic solid crusts or cores. (Schumaker & Thorne 1983 have presented a similar treatment for torsional modes.) The pulsations are treated as first-order perturbations of non-rotating perfect fluid spheres, with an isotropic shear modulus describing a linear relationship between the shear strain and stress. The Einstein Field Equations (EFE), equations of motion and boundary conditions describing the quasi-normal mode pulsations of these stars are found, and are in turn used to find a Lagrangian density and the corresponding variational principle for the system of pulsational modes.

The pulsational equations for these stars form a sixth-order system, corresponding to the three broad types of modes that may be excited. One set of modes may be characterized as *fluid pulsational modes*, for which the fluid motion is large and the gravitational perturbation is small; a second set of modes may be characterized as *gravitational modes*, for which the perturbation of the gravitational field is large and the fluid perturbation is small; and a third set of modes corresponds to *elastic modes* which are confined predominantly to the elastic crust and also have small gravitational perturbations.

The gravitational modes were first discussed by Dyson (1980) and later by Kokkotas & Schutz (1986) in the context of model problems. Kojima (1988) has described and implemented a technique for calculating these modes. Gravitational pulsational modes are not likely to have significant astrophysical importance.

In fact these modes are all coupled; however, in practice the coupling between the gravitational and fluid or elastic modes is usually weak. Thus, it can be fruitful to investigate the fluid and elastic modes by discarding the gravitational perturbations entirely in a *Cowling approximation* (Cowling 1941). For this purpose, McDermott, Van Horn & Scholl (1983) introduced a Cowling approximation for relativistic stars, and Finn (1988) suggested a modification of their approximation and used it to prove several theorems regarding the fluid modes. Recently, Lindblom & Splinter (1989a) have investigated numerically the behaviour of these two relativistic Cowling approximations for p-modes (see below). McDermott and co-workers have investigated fluid and elastic modes in great detail in the context of their relativistic Cowling approximation (McDermott *et al.* 1983, 1985; McDermott 1985; McDermott, Van Horn & Hansen 1988).

In order to investigate the coupling between the gravitational modes and the fluid/elastic modes, or to understand in detail how gravitational radiation is generated by stellar pulsations, a Cowling approximation is insufficient. In these cases, a complete treatment that encompasses all the modes without approximation is necessary. Such a general perturbation theory was developed for radial modes by Chandrasekhar (1964), and for non-radial modes in perfect fluid stars without an elastic modulus by Thorne, Campolattaro, Price, Ipser and Detweiler (Thorne & Campolattaro 1967a,b; Price & Thorne 1969; Thorne 1969a,b; Campolattaro & Thorne 1970; Ipser & Thorne 1973; Detweiler & Ipser 1973; Detweiler 1975a,b). Important recent contributions to the theory of p-mode and f-mode pulsations have been made by Detweiler & Lindblom (1985) and Lindblom & Detweiler (1983), and to the theory of g-modes by McDermott *et al.* (1983), Finn (1986, 1987, 1988) and Lindblom & Splinter (1989a,b). Schumaker & Thorne (1983) extended this theory to encompass the odd-parity, torsional modes introduced by the presence of a solid crust. The goal of this paper is to extend the theory of non-radial pulsations to encompass stars with an isotropic shear modulus.

In addition to introducing new modes, the presence of an elastic modulus will modify a neutron star's existing modes. Non-radial fluid modes of perfect fluid stars are generally divided into three classes: p-modes, f-modes and g-modes. The p-modes are essentially sound waves, with predominantly longitudinal fluid motion supported by pressure. The f-modes, which exist only for $l \geq 2$, have much of the character of surface gravity waves while the g-modes, which exist only for $l \geq 1$, are subsurface gravity waves. For the g-modes, the fluid motion is primarily transverse with a large shear component. At fixed spherical harmonic order $l \geq 2$ and $|m| \leq l$, there is a single f-mode, for $l \geq 1$ a semi-infinite spectrum of g-modes, and for $l \geq 0$ a semi-infinite spectrum of p-modes. Since the restoring force for g-modes is weak and the shear is large, we expect that the new elastic modes will have their greatest effect upon the g-mode spectrum.

The paper is organized as follows: the remainder of Section 1 introduces the notation and conventions used throughout the rest of the paper. In Section 2 the shear strain and stress are derived in terms of the Thorne–Campolattaro perturbation variables. Section 3 presents the EFE, equations of motion and boundary conditions for the non-radial pulsational modes. In Section 4, a Lagrangian density describing the pulsations is derived from the field equations and equations of motion, and the corresponding variational principle, which can be used to investigate the eigenfunctions and eigenfrequencies, is stated. Conclusions are presented in Section 5.

1.2 Conventions and notation

Lower case Greek indices run over space-time coordinates and lower case Latin indices run over spatial coordinates. A semi-colon signals a covariant derivative in space-time, while a comma represents an affine (coordinate) derivative. We also use the conventional shorthand notation for derivatives with respect to coordinate radius r and t

$$f' \equiv \frac{\partial f}{\partial r} = f_{,r}, \tag{1}$$

$$\dot{f} \equiv \frac{\partial f}{\partial t} = f_{,t}. \tag{2}$$

Upper case Latin indices run over the coordinates θ and ϕ on the constant r 2-sphere with negative-definite line element

$$ds^2 = -r^2(\mathrm{d}\theta^2 + \sin^2\theta \mathrm{d}\phi^2), \tag{3}$$

and a vertical bar ($|$) is used to represent the covariant derivative on this 2-sphere.

When discussing boundary conditions, we will have occasion to refer to the discontinuity in some function. We adopt the notation $[f]_r$ to refer to the discontinuity in the quantity f across the radial shell at r:

$$[f]_r \equiv \lim_{\varepsilon \to 0+} f(r+\varepsilon) - f(r-\varepsilon). \tag{4}$$

The theory of non-radial stellar pulsations is formulated as a general perturbation about an exact solution of the EFE. The exact solution is described by the line element

$$\mathrm{d}s_0^2 = e^{\nu}\mathrm{d}t^2 - e^{\lambda}\mathrm{d}r^2 - r^2(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2), \tag{5}$$

where $\nu(r)$ and $\lambda(r)$ are defined by the relativistic equations of stellar structure:

$$\frac{d\nu}{dr} = \frac{2(m + 4\pi r^3 P)}{r(r-2m)}, \tag{6a}$$

$$e^{-\lambda} = 1 - \frac{2m}{r}, \tag{6b}$$

$$\frac{dP}{dr} = -\frac{\rho + P}{2}\frac{d\nu}{dr}, \tag{6c}$$

$$m(r) = \int_0^r 4\pi r^2 \rho dr. \tag{6d}$$

The perturbations are resolved into spherical harmonics (Y_{lm}) and exponential time-dependence $\exp(-i\omega t)$ with complex frequency $\omega \equiv \sigma - i/\tau$ (σ and τ real). The metric perturbations are expressed in Regge–Wheeler gauge (Regge & Wheeler 1957): for spherical harmonic order l and m, four functions (H_0, $\dot{H}_1$, H_2 and K) describe the perturbed metric:

$$\mathrm{d}s^2 \equiv \mathrm{d}s_0^2 + (e^{\nu}H_0\mathrm{d}t^2 + 2\dot{H}_1\mathrm{d}t\,\mathrm{d}r + e^{\lambda}H_2\,\mathrm{d}r^2 + r^2K\,\mathrm{d}\Omega^2)\,Y_{lm}. \tag{7}$$

With the exception of $\dot{H}_1$, these are the same definitions as in Thorne & Campolattaro (1967a). In most applications it is more convenient to work with H_1,

$$H_1 \equiv \int dt\,\dot{H}_1, \tag{8}$$

and the recent literature uses this definition.

The fluid degrees of freedom are expressed in terms of the radial dependence of the radial and angular coordinate displacement of a fluid element from its equilibrium location. Denoting the coordinate displacement of a fluid element from its equilibrium location as ξ, the fluid perturbations are W and V, where

$$\xi^r \equiv r^{-2}e^{-\lambda/2}\,W Y_{lm}, \tag{9}$$

$$\xi_\theta \equiv V\frac{\partial Y_{lm}}{\partial\theta}, \tag{10}$$

$$\xi_\phi \equiv V\frac{\partial Y_{lm}}{\partial\phi}. \tag{11}$$

Like the gravitational perturbations H_0, $\dot{H}_1$, H_2 and K, the fluid displacement functions W and V are functions of r and t only.

2 SHEAR STRAIN AND STRESS

Carter & Quintana (1972) extended the standard theory of elastic solids to general relativity. In their theory of elasticity, the shear $\Sigma_{\mu\nu}$ is derived from the rate of shear $\sigma_{\mu\nu}$ and the rate of expansion θ through

$$[\Sigma_{\mu\nu}]^{\cdot} = \tfrac{2}{3}\,\theta\Sigma_{\mu\nu} + \sigma_{\mu\nu}, \tag{12}$$

where $[\;]^{\cdot}$ is the proper time derivative co-moving with the shearing matter (the *convected time derivative* of Carter & Quintana 1972). The rate of shear and expansion are themselves defined in terms of the fluid 4-velocity $\mathbf{U}$:

$$\theta \equiv \perp_{\mu\nu}\mathbf{U}^{\mu;\nu} = \mathbf{U}^{\mu}_{;\mu}, \tag{13a}$$

$$\sigma_{\mu\nu} \equiv \tfrac{1}{2}\left(\perp^{\alpha}_{\mu}\mathbf{U}_{\nu;\alpha} + \perp^{\alpha}_{\nu}\mathbf{U}_{\mu;\alpha}\right) - \tfrac{1}{3}\perp_{\mu\nu}\theta, \tag{13b}$$

where $\perp^{\nu}_{\mu}$ is the projection tensor orthogonal to $\mathbf{U}$

$$\perp \equiv \mathbf{g} - \mathbf{U}\otimes\mathbf{U}. \tag{14}$$

The shear and the rate of shear are both orthogonal to the fluid 4-velocity, and for the fully covariant tensor $\Sigma_{\mu\nu}$ the convected time derivative reduces to the Lie derivative along $\mathbf{U}$.

We treat stellar pulsations as a first-order perturbation upon a static, strain-free equilibrium background. Consequently, the strain is of order the perturbation and the term $\theta\Sigma$ in equation (12) is negligible. Similarly, the deviations of the fluid 4-velocity from rest in the equilibrium Schwarzschild space-time are of the order of the perturbation, and the convected time derivative in equation (12) reduces to

$$[\;]^{\cdot} = \frac{d}{d\tau} = e^{-\nu/2}\frac{d}{dt}. \tag{15}$$

Consequently, the relation between the σ and Σ takes on the especially simple form

$$\Sigma_{\mu\nu} = e^{\nu/2}\int_0^t dt\,\sigma_{\mu\nu}. \tag{16}$$

The perturbation is assumed to vanish at $t=0$.

We do not consider here the bulk compressional strain and related stress. That stress would enter the equations exactly as an additional component of the pressure and could be included in the fluid equation of state.

Our assumption of vanishing background strain is almost certainly incorrect for realistic stars. Nevertheless, the background shear should still be only a small contribution to the total stress-energy of the equilibrium configuration. In the limit of small perturbations, but smaller background shear, our approximation of vanishing background shear ought to be valid.

The 4-velocity of the non-radially pulsating material is given by

$$\mathbf{U}^t = e^{-\nu/2}\left(1 - \tfrac{1}{2}H_0 Y_{lm}\right), \tag{17a}$$

$$\mathbf{U}^r = \frac{e^{-(\nu+\lambda)/2}}{r^2}\,\dot{W}Y_{lm}, \tag{17b}$$

$$\mathbf{U}_\theta = e^{-\nu/2}\,\dot{V}Y_{lm,\theta}, \tag{17c}$$

$$\mathbf{U}_\phi = e^{-\nu/2}\,\dot{V}Y_{lm,\phi}. \tag{17d}$$

From $\mathbf{U}$ and equations (13a,b) and (16), it is straightforward to calculate the shear strain tensor:

$$\Sigma^t_\mu = 0, \tag{18a}$$

$$\Sigma^r_r \equiv \mathscr{A}\,Y_{lm} = \frac{1}{3}\left[K - H_2 + 2re^{-\lambda/2}\left(\frac{W}{r^3}\right)' - \frac{l(l+1)}{r^2}V\right]Y_{lm}, \tag{18b}$$

$$\Sigma^r_A \equiv \mathscr{B}\,Y_{lm,A} = \frac{1}{2}\left[\frac{e^{-\lambda/2}}{r^2}W - r^2e^{-\lambda}\left(\frac{V}{r^2}\right)'\right]Y_{lm,A}, \tag{18c}$$

$$\begin{aligned}\Sigma^A_B &\equiv \mathscr{C}\,Y_{lm}{}_B^{|A} + \mathscr{D}\,\delta^A_B Y_{lm}\\ &= VY_{lm}{}^{|A}{}_B\\ &\quad + \frac{1}{6}\left[H_2 - K - 2re^{-\lambda/2}\left(\frac{W}{r^3}\right)' - 2\frac{l(l+1)}{r^2}V\right]\delta^A_B Y_{lm}.\end{aligned} \tag{18d}$$

Recall that the uppercase Latin indices A and B run over the coordinates θ and ϕ, the vertical bar (as in $Y_{lm}{}^{|A}{}_B$) indicates a covariant derivative on the 2-sphere with *negative definite* line element $ds^2 = -r^2(d\theta^2 + \sin^2\theta\,d\phi^2)$, and δ^A_B is the Kronecker delta. The remaining elements of $\Sigma_{\mu\nu}$ can be found by raising and lowering indices with the projection tensor $\perp_{\mu\nu}$ (*cf.* equation 14). Note that the trace (with respect to $\perp^{\nu}_{\mu}$) of Σ^{μ}_{ν} vanishes:

$$\perp^{\nu}_{\mu}\Sigma^{\mu}_{\nu} = \mathscr{A} + \frac{l(l+1)}{r^2}\mathscr{C} + 2\mathscr{D} = 0. \tag{19}$$

We assume a Hookean relationship between the shear strain and stress,

$$\delta\mathbf{T}^{\text{shear}} = 2\mu\boldsymbol{\Sigma}, \tag{20}$$

where μ is the isotropic shear modulus. The total stress energy tensor is thus

$$\mathbf{T} \equiv (\rho + P)\mathbf{U}\otimes\mathbf{U} - P\mathbf{g} + 2\mu\boldsymbol{\Sigma}. \tag{21}$$

3 THE EIGENVALUE PROBLEM FOR NON-RADIAL NORMAL MODE PULSATIONS

3.1 Field equations

Thorne & Zimmerman (1967) give the perturbed EFE in the Regge–Wheeler (Regge & Wheeler 1957) gauge for non-radial pulsations. Using their results together with the stress-energy tensor of a perturbed perfect fluid (Thorne & Campolattaro 1967a) and our own results for the shear stress of the solid crust (equation 20), we can write down the EFE for the perturbation of a spherical star with an isotropic shear modulus:

$r^2[^t_t]$:

$$\begin{aligned}&e^{-\lambda}r^2K'' + (3 - r\lambda'/2)e^{-\lambda}rK' - \frac{(l+2)(l-1)}{2}K - e^{-\lambda}rH_2'\\ &\quad - \left[e^{-\lambda}(1 - r\lambda') + \frac{l(l+1)}{2}\right]H_2 = 8\pi r^2\alpha,\end{aligned} \tag{22}$$

$2r^2 e^{\lambda}[{}^{r}_{t}]$:

$$l(l+1)\dot{H}_1 + 2r[H_{2,t} - e^{\nu/2}(e^{-\nu/2} rK_{,t})'] = 16\pi(\rho+P)e^{\lambda/2} W_{,t}, \tag{23}$$

$2r^2 e^{(\nu+\lambda)/2}[{}^{A}_{t}]$:

$$[e^{(\nu-\lambda)/2}\dot{H}_1]' - e^{(\nu+\lambda)/2}(K_{,t} + H_{2,t}) = -16\pi e^{(\nu+\lambda)/2}(\rho+P)V_{,t}, \tag{24}$$

$r^2[{}^{r}_{r}]$:

$$\begin{aligned}(1+r\nu'/2)e^{-\lambda} rK' - e^{-\nu} r^2 K_{,tt} - \frac{(l+2)(l-1)}{2}K \\ - e^{-\lambda}(1+r\nu')H_2 + \frac{l(l+1)}{2}H_0 - e^{-\lambda} rH_0' \\ + 2re^{-(\nu+\lambda)}\dot{H}_{1,t} = -8\pi r^2(\beta - 2\mu\mathscr{A}),\end{aligned} \tag{25}$$

$2r^2[{}^{A}_{r}]$:

$$\begin{aligned}r^{-1}(1+r\nu'/2)H_2 - r^{-1}(1-r\nu'/2)H_0 \\ + H_0' - K' - e^{-\nu}\dot{H}_{1,t} = 32\pi\mu e^{\lambda}\mathscr{B},\end{aligned} \tag{26}$$

$2[{}^{A}_{B}]$:

$$H_0 - H_2 = -32\pi\mu\mathscr{C}, \tag{27}$$

$2e^{-\lambda}[{}^{A}_{B}]$:

$$\begin{aligned}K'' - H_0'' + 2e^{-\nu+\lambda/2} r^{-1}[e^{-\lambda/2} r\dot{H}_{1,t}]' \\ - r^{-1}[1 + r(\nu' - \lambda'/2)]H_0' - r^{-1}(1+r\nu'/2)H_2' \\ + r^{-1}[2 + r(\nu' - \lambda')/2]K' - e^{\lambda-\nu}(K+H_2)_{,tt} \\ + r^{-2}\{r\lambda' + r\nu'[r(\lambda' - \nu')/2 - 1] - r^2\nu''\}H_2 \\ - e^{\lambda}\frac{l(l+1)}{r^2}(H_2 - H_0) = -16\pi e^{\lambda}(\beta - 2\mu\mathscr{D}).\end{aligned} \tag{28}$$

Equations (27) and (28) correspond to different types of spherical harmonic dependence: the angular dependence factored out of equation (27) is $Y_{lm}{}^{;A}{}_{B}$, and the dependence factored out of equation (28) is Y_{lm}. Note that we have used Regge–Wheeler (1957) spherical harmonics to perform the angular decomposition. The primary difference between this basis and the more conventional tensor spherical harmonic bases (*cf.* Thorne 1980) is that the Regge–Wheeler basis is not orthonormal. For this particular problem, however, it is more convenient to use the Regge–Wheeler basis, and we will continue to do so. The notation $[{}^{\mu}_{\nu}]$ indicates the radially dependent part of the field equation $\delta G^{\mu}_{\nu} = 8\pi\delta T^{\mu}_{\nu}$ (where $\delta\mathbf{G}$ is the perturbed Einstein tensor and $\delta\mathbf{T}$ the perturbed stress-energy tensor). For notational convenience, the radial dependence of the Eulerian pressure perturbation is abbreviated $\beta(\delta P \equiv \beta Y_{lm})$.

The continuity equation $[\nabla\cdot(n\mathbf{U}) = 0$, where n is the baryon number density] determines the relationship between β and H_0, $\dot{H}_1$, H_2, K, W and V:

$$\beta = \gamma P\left[\frac{1}{2}H_2 + K - \frac{l(l+1)}{r^2}V - \frac{e^{-\lambda/2}}{r^2}\left(W' + \frac{P'}{\gamma P}W\right)\right], \tag{29}$$

where γ relates the Lagrangian perturbations in pressure (ΔP) and baryon number density (Δn) through

$$\frac{\gamma P}{\Delta P} \equiv \frac{n}{\Delta n} \tag{30}$$

(definition of γ). Also for notational convenience, the radial dependence of the Eulerian density perturbation $\delta\rho$ is denoted by $\alpha(\delta\rho \equiv \alpha Y_{lm})$. Since the Lagrangian density perturbation $(\Delta\rho)$ is related to the Lagrangian pressure perturbation through

$$\frac{\gamma P}{\Delta P} = \frac{n}{\Delta n} = \frac{\rho+P}{\Delta\rho}, \tag{31}$$

α can be expressed in terms of the metric and fluid perturbation variables:

$$\begin{aligned}\alpha &= \frac{\rho+P}{\gamma P}\left(\beta + \frac{e^{-\lambda/2}}{r^2}P'W\right) - \frac{e^{-\lambda/2}}{r^2}\frac{d\rho}{dr}W \\ &= \frac{\rho+P}{\gamma P}\beta + \frac{e^{-\lambda/2}}{r^2}\frac{d\rho}{dr}\left(\frac{\gamma_0}{\gamma} - 1\right)W.\end{aligned} \tag{32}$$

Here γ_0 denotes the effective adiabatic index of the equilibrium equation of state:

$$\gamma_0 \equiv \frac{dP/dr}{d\rho/dr}\frac{\rho+P}{P}. \tag{33}$$

We distinguish between γ and γ_0 because γ_0 may reflect an equilibrium, or the completion of chemical or nuclear processes, that takes place on time-scales which are long compared to the pulsational time-scale. For example, the pulsational period may be short enough that the perturbation is adiabatic while the neutron star itself may be (in some region) isothermal.

3.2 Equations of motion

Thorne & Zimmerman (1967) also give the divergence of the stress-energy tensor for a general even-parity perturbation of a perfect fluid. Together with the divergence of equation (20) this gives the equations of motion for fluid element displacement in the solid crust star:

$$\begin{aligned}\delta\mathbf{T}_r:\ -e^{-\nu/2}(e^{\nu/2}\beta)' + \frac{P'}{\gamma P}\beta - e^{-\nu}(\rho+P)\left[\frac{e^{\lambda/2}}{r^2}W_{,tt} - \dot{H}_{1,t}\right] \\ - \frac{1}{2}(\rho+P)H_0' + \frac{P'}{\gamma P}\frac{e^{-\lambda/2}}{r^2}WS - 2\mu e^{\lambda}\frac{l(l+1)}{r^2}\mathscr{B} \\ + \frac{2\mu}{r}\mathscr{A} + 2\frac{(\mu r^2 e^{\nu/2}\mathscr{A})'}{r^2 e^{\nu/2}} = 0,\end{aligned} \tag{34}$$

$$\begin{aligned}\delta\mathbf{T}_A:\ -\beta - \omega^2 e^{-\nu}(\rho+P)V - \frac{1}{2}(\rho+P)H_0 + 2\mu\mathscr{D} \\ + \frac{2\mu}{r^2}[l(l+1) - 1]\mathscr{C} + 2\frac{[\mu r^2 e^{(\nu+\lambda)/2}\mathscr{B}]'}{r^2 e^{(\nu+\lambda)/2}} = 0,\end{aligned} \tag{35}$$

where S is the relativistic Schwarzschild discriminant

$$S \equiv P' - \frac{\gamma P}{\rho + P}\rho', \tag{36}$$

and $\delta \mathrm{T}_\mu$ represents the equation of motion $\delta \mathrm{T}^{\nu}_{\mu;\nu}$ with the spherical harmonic dependence (Y_{lm} for equation 34, $Y_{lm;A}$ for equation 35) factored out. Equation (29) has been used to simplify the Euler equations. The perturbation of the equation of energy conservation, $\mathrm{T}^{t\alpha}_{;\alpha} = 0$, vanishes identically.

3.3 A complete set of equations

Not all of the nine equations (22)–(28) and (34)–(35) are independent. When μ vanishes throughout the star, they reduce to a fourth-order system with two algebraic constraints (*cf.* Ipser & Thorne 1973). In the case considered here, $\mu \neq 0$ and the star can support shear modes. This introduces a new degree of freedom and the equations now form a sixth-order system. In this subsection, we find a complete subset of the EFE and equations of motion.

Since (23)–(26) all are independent (by inspection) first-order ordinary differential equations, they form a nice starting place for establishing a complete set of independent equations. The pair of equations (23) and (24) include the information available in (22) (through the Bianchi identity $\mathbf{G}^{\mu}_{t;\mu} = 0$); similarly, the triplet of equations (25), (23) and (26) make equation (34) redundant (through $\mathbf{G}^{\mu}_{r;\mu} = 0$ and $\mathbf{T}^{\mu}_{r;\mu} = 0$). There are only two remaining equations that we can choose from to complete our set: (35) and (28) (with either of these we are also allowed 27). Choosing either one makes the other redundant through the use of (24), (26) and the Bianchi identity $\mathbf{G}^{\mu}_{A;\mu} = 0$ or (equivalently) the equation of motion $\mathbf{T}^{\mu}_{A;\mu} = 0$.

When $\mu = 0$, equation (35) reduces to an algebraic equation; thus it is a particularly compelling choice for our final equation. When $\mu \neq 0$, equation (35) is a second-order differential equation, the second-order term being $\mu V''$. This equation is singular when $\mu \to 0$ as $r \to r_i$, and we must be careful how we integrate the system in this case. Regardless, the complete system of equations describing the stellar pulsations is

$$l(l+1)\dot{H}_1 + 2r[H_{2,t} - e^{\nu/2}(e^{-\nu/2} rK_{,t})'] = 16\pi(\rho + P)e^{\lambda/2} W_{,t}, \tag{37}$$

$$[e^{(\nu-\lambda)/2}\dot{H}_1]' - e^{(\nu+\lambda)/2}(K_{,t} + H_{2,t}) = -16\pi e^{(\nu+\lambda)/2}(\rho + P)V_{,t}, \tag{38}$$

$$(1 + r\nu'/2)e^{-\lambda} rK' - e^{-\nu} r^2 K_{,tt} - \frac{(l+2)(l-1)}{2}K - e^{-\lambda}(1 + r\nu')H_2 + \frac{l(l+1)}{2}H_0 - e^{-\lambda} rH_0' + 2re^{-(\nu+\lambda)}\dot{H}_{1,t} = -8\pi r^2(\beta - 2\mu\mathscr{A}), \tag{39}$$

$$r^{-1}(1 + r\nu'/2)H_2 - r^{-1}(1 - r\nu'/2)H_0 + H_0' - K' - e^{-\nu}\dot{H}_{1,t} = 32\pi\mu e^{\lambda}\mathscr{B}, \tag{40}$$

$$H_0 - H_2 = -32\pi\mu\mathscr{C}, \tag{41}$$

$$-\beta - \omega^2 e^{-\nu}(\rho + P)V - \frac{1}{2}(\rho + P)H_0 + 2\mu\mathscr{D} + \frac{2\mu}{r^2}[l(l+1) - 1]\mathscr{C} + 2\frac{[\mu r^2 e^{(\nu+\lambda)/2}\mathscr{B}]'}{r^2 e^{(\nu+\lambda)/2}} = 0. \tag{42}$$

3.4 Boundary conditions

In the previous section we gave a complete set of differential and algebraic equations for the perturbation of the star. In this subsection, we impose boundary conditions appropriate for the quasi-normal (damped and undriven) pulsational modes on those equations. Together, these boundary conditions and the complete set of equations from the previous section form an eigenvalue problem for the complex pulsational frequency ω.

There are at least three places where boundary conditions must be imposed: at the star's centre, at its surface and at infinity. Additionally, we must impose boundary conditions anywhere the shear modulus is discontinuous or vanishes (i.e. at any crustal interfaces). Boundary conditions must also be imposed at discontinuities in density, such as are encountered in the Baym, Pethick & Sutherland (1971) equation of state (*cf.* Finn 1987). In what follows, we assume that the density is continuous everywhere except perhaps at crustal interfaces. Finn (1987) describes how to treat isolated density discontinuities.

3.4.1 *Boundary conditions at the star's centre*

At the centre of the star, the perturbations must be regular. Together with the tensorial character of the perturbation variables, this leads to conditions on the radial dependence of the perturbation variables in the neighbourhood of the origin:

$$H_0 = r^l[H_0^{(0)} + H_0^{(2)} r^2 + \ldots], \tag{43a}$$

$$H_1 = r^{l+1}[H_1^{(0)} + H_1^{(2)} r^2 + \ldots], \tag{43b}$$

$$H_2 = r^l[H_2^{(0)} + H_2^{(2)} r^2 + \ldots], \tag{43c}$$

$$K = r^l[K^{(0)} + K^{(2)} r^2 + \ldots], \tag{43d}$$

$$W = r^{l+1}[W^{(0)} + W^{(2)} r^2 + \ldots], \tag{43e}$$

$$V = r^l[V^{(0)} + V^{(2)} r^2 + \ldots]. \tag{43f}$$

Additionally, the Eulerian pressure perturbation is regular, so that

$$\beta = r^l[\beta^{(0)} + \beta^{(2)} r^2 + \ldots]. \tag{44}$$

Finally, the shear *stress* must be regular at the origin:

$$\mu\mathscr{A} = r^l\{[\mu\mathscr{A}]^{(0)} + [\mu\mathscr{A}]^{(2)} r^2 + \ldots\}, \tag{45a}$$

$$\mu\mathscr{B} = r^{l+1}\{(\mu\mathscr{B})^{(0)} + (\mu\mathscr{B})^{(2)} r^2 + \ldots\}, \tag{45b}$$

$$\mu\mathscr{C} = r^l\{[\mu\mathscr{C}]^{(0)} + [\mu\mathscr{C}]^{(2)} r^2 + \ldots\}, \tag{45c}$$

$$\mu\mathscr{D} = r^l\{(\mu\mathscr{D})^{(0)} + (\mu\mathscr{D})^{(2)} r^2 + \ldots\}. \tag{45d}$$

The character of these solutions is determined by the transformation properties of the perturbation variables under rotations: H_0, H_2, K, β, $\mu\mathscr{A}$, $\mu\mathscr{C}$ and $\mu\mathscr{D}$ transform as scalars under rotations, W, H_1 and $\mu\mathscr{B}$ transform as the radial derivatives of scalars, and V transforms as a vector on the sphere.

Not all of these conditions are independent; in fact, we have already expressed $\beta, \mathscr{A}, \mathscr{B}, \mathscr{C}$ and $\mathscr{D}$ in terms of W, V, H_0, H_1, H_2 and K (*cf.* equations 29 and 18a–d). The definition of β (equation 29) leads to

$$W^{(0)} = -lV^{(0)}. \tag{46}$$

Additionally, the definition of the strain components $\mathscr{A}$, $\mathscr{B}$ and $\mathscr{D}$ (equations 18b–d) require that

$$\mu \neq \mathscr{O}(r^2) \rightarrow W^{(0)} = V^{(0)} = 0. \tag{47}$$

This suppression of $W^{(0)}$ and $V^{(0)}$ when $\mu \neq \mathscr{O}(r^2)$ is not unexpected. The rigidity of the star at the origin, reflected in the shear modulus, requires that the strain be smoother than would otherwise be the case. The situation is analogous to one that occurs when the Euler and Navier–Stokes equations are compared: in the former case, there is no boundary condition on the fluid's tangential velocity at a boundary, while in the latter case the tangential velocity must vanish smoothly.

Once regularity of the solutions is established, the perturbation equations lead to relations between the coefficients $H_0^{(0)}$, etc., of the gravitational perturbations and those of the fluid perturbations;

$$K^{(0)} = H_2^{(0)} = H_0^{(0)}, \tag{48a}$$

$$H_1^{(0)} = (l+1)^{-1}[2H_0^{(0)} - 16\pi(\rho+P)V^{(0)}], \tag{48b}$$

$$\frac{1}{2}[(3\gamma+1)P+\rho]H_0^{(0)} = \left[\gamma P(l+3) + \frac{\mu}{3}(l+9)\right]W^{(2)} + \left[\gamma Pl + \frac{\mu}{3}(l-6)\right]V^{(2)}. \tag{48c}$$

Equation (48a) comes from (41) and (40), (48b) comes from (38), and (48c) comes from (42). In finding (48c), we have assumed

$$\mu = \mu^{(0)} + \mu^{(2)}r^2 + \ldots \tag{49}$$

in a neighbourhood of the origin. Thus regularity of the perturbation restricts us to only two solutions to the perturbation equations near the origin, corresponding to an arbitrary choice of, e.g. $H_0^{(0)}$ and either $V^{(0)}$ [if $\mu \neq \mathscr{O}(r^2)$] or $V^{(2)}$.

3.4.2 *Crustal interfaces*

The crust in a neutron star may occupy one or more spherical shells. At the boundary of these shells the shear modulus changes discontinuously. This may occur within a single elastic solid, or at the interface between an elastic solid and a fluid. In the former case, the shear modulus discontinuously changes but does not vanish on either side of the boundary, while in the latter it discontinuously vanishes. In either case, we must impose boundary conditions on the shell where the shear modulus changes.

The boundary conditions on these shells arise from the equations of motion and the EFE, as well as the character of the fluid and the elastic solid. The equations of motion require continuity of the stress across the (perturbed) shell, while the EFE require continuity of the intrinsic geometry and the extrinsic curvature there. In addition, at a boundary within a single elastic solid, the displacement must also be continuous. At the interface between an elastic solid and an ideal fluid, however, only the normal displacement is continuous: the fluid can slip freely along the interface.

In order to develop the boundary conditions on the stress, intrinsic geometry, and extrinsic curvature, we need to locate and describe the perturbed interface. The interface is located at $r = r_i + \xi^r$, where r_i is the radial coordinate of the shell in the unperturbed star (or, alternatively, the Lagrangian coordinate radius of the interface). Hence, it is described by the one-form

$$\begin{aligned}\mathbf{n}_\mu \mathbf{d}x^\mu &= \mathbf{d}(r-\xi^r)\\ &= -\frac{e^{-\lambda/2}}{r^2}\dot{W}Y_{lm}\mathbf{d}t + \left[1 - \left(\frac{e^{-\lambda/2}}{r^2}WY_{lm}\right)'\right]\mathbf{d}r\\ &\quad - \frac{e^{-\lambda/2}}{r^2}W(Y_{lm,\theta}\mathbf{d}\theta + Y_{lm,\phi}\mathbf{d}\phi).\end{aligned} \tag{50}$$

The fully contravariant form of the four-metric is

$$\begin{aligned}\mathbf{g}^{\mu\nu}\frac{\partial}{\partial x^\mu}\frac{\partial}{\partial x^\nu} &= e^{-\nu}(1 - H_0Y_{lm})\frac{\partial}{\partial t}\frac{\partial}{\partial t} + 2e^{-(\lambda+\nu)}H_1Y_{lm}\frac{\partial}{\partial t}\frac{\partial}{\partial r}\\ &\quad - e^{-\lambda}(1+H_2Y_{lm})\frac{\partial}{\partial r}\frac{\partial}{\partial r}\\ &\quad - r^{-2}(1+KY_{lm})\left(\frac{\partial}{\partial\theta}\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial}{\partial\phi}\frac{\partial}{\partial\phi}\right);\end{aligned} \tag{51}$$

so the unit one-form describing the surface is

$$\mathbf{N}_\mu \mathbf{d}x^\mu = e^{\lambda/2}\{1 + [(r^{-2}e^{-\lambda/2}W)' - \tfrac{1}{2}H_2]Y_{lm}\}\mathbf{n}_\mu\mathbf{d}x^\mu \tag{52}$$

$$\begin{aligned}&= -r^{-2}\dot{W}Y_{lm}\mathbf{d}t + e^{\lambda/2}[1 - \tfrac{1}{2}H_2Y_{lm}]\mathbf{d}r\\ &\quad - r^{-2}W(Y_{lm,\theta}\mathbf{d}\theta + Y_{lm,\phi}\mathbf{d}\phi),\end{aligned} \tag{53}$$

and the corresponding unit normal is

$$\begin{aligned}\mathbf{N}^\mu\frac{\partial}{\partial x^\mu} &= e^{-\nu}(e^{-\lambda/2}H_1 - r^{-2}\dot{W})Y_{lm}\frac{\partial}{\partial t} - e^{-\lambda/2}\left(1 + \frac{1}{2}H_2Y_{lm}\right)\\ &\quad \times\frac{\partial}{\partial r} + r^{-4}W\left(Y_{lm,\theta}\frac{\partial}{\partial\theta} + \frac{1}{1\sin^2\theta}Y_{lm,\phi}\frac{\partial}{\partial\phi}\right).\end{aligned} \tag{54}$$

The surface itself is spanned by the unit vectors

$$\mathbf{e}_{\hat{t}} \equiv e^{-\nu/2}\left(1 - \frac{1}{2}H_0Y_{lm}\right)\frac{\partial}{\partial t} + \frac{e^{-(\nu+\lambda)/2}}{r^2}\dot{W}Y_{lm}\frac{\partial}{\partial r}, \tag{55a}$$

$$\mathbf{e}_{\hat{\theta}} \equiv r^{-1}\left(1 + \frac{1}{2}KY_{lm}\right)\frac{\partial}{\partial\theta} + \frac{e^{-\lambda/2}}{r^3}WY_{lm,\theta}\frac{\partial}{\partial r}, \tag{55b}$$

$$\mathbf{e}_{\hat{\phi}} = \frac{1}{r\sin\theta}\left(1 + \frac{1}{2}KY_{lm}\right)\frac{\partial}{\partial\phi} + \frac{e^{-\lambda/2}}{r^3}WY_{lm,\phi}\frac{\partial}{\partial r}. \tag{55c}$$

The projection of the stress tensor normal to the perturbed surface is

$$\begin{aligned}\perp\cdot\mathbf{T}\cdot\perp\cdot\mathbf{N}|_{r_i+\xi^r} &= -[(P+\beta Y_{lm})(\mathbf{N} - \mathbf{U}\otimes\mathbf{U}\cdot\mathbf{N})\\ &\quad - 2\mu\mathbf{\Sigma}\cdot\mathbf{N}]|_{r_i+\xi^r}\\ &= -[(P+\beta Y_{lm})\mathbf{N} - 2\mu\mathbf{\Sigma}\cdot\mathbf{N}]|_{r_i+\xi^r},\end{aligned} \tag{56}$$

or

$$\mathbf{e}_{\hat{t}}\cdot\perp\cdot\mathbf{T}\cdot\perp\cdot\mathbf{N} = 0 \tag{57a}$$

$$\mathbf{N}\cdot\perp\cdot\mathbf{T}\cdot\perp\cdot\mathbf{N} = (P+\beta Y_{lm}) - 2\mu\mathscr{A}Y_{lm}, \tag{57b}$$

$$\mathbf{e}_{\hat{\theta}}\cdot\perp\cdot\mathbf{T}\cdot\perp\cdot\mathbf{N} = -2r^{-1}\mu e^{\lambda/2}\mathscr{B}Y_{lm,\theta}. \tag{57c}$$

Each of these is continuous across the perturbed interface. This yields two conditions:

$$0=\left[\beta+\frac{e^{-\lambda/2}}{r^2}P'W-2\mu\mathscr{A}\right]_{r_i}$$
$$=\left\{\gamma P\left[\frac{1}{2}H_2+K-\frac{l(l+1)}{r^2}V-\frac{e^{-\lambda/2}}{r^2}W'\right]\right.$$
$$\left.-\frac{2\mu}{3}\left[K-H_2-\frac{l(l+1)}{r^2}V\right]-\frac{4\mu}{3}re^{-\lambda/2}\left(\frac{W}{r^3}\right)'\right\}_{r_i} \tag{58a}$$

$$0=[2e^{\lambda/2}\mu\mathscr{B}]_{r_i}$$
$$=\left[\frac{\mu}{r^2}W-\mu r^2e^{-\lambda/2}\left(\frac{V}{r^2}\right)'\right]_{r_i}. \tag{58b}$$

Condition (58a) is essentially continuity of the Lagrangian pressure perturbation, adjusted for the new elastic stress.

The intrinsic geometry is described by the three-metric induced on the perturbed surface:

$$\begin{aligned}{}^{(3)}\mathbf{g}_{\mu\nu}\mathbf{d}x^\mu\mathbf{d}x^\nu&=(\mathbf{g}_{\mu\nu}+\mathbf{N}_\mu\mathbf{N}_\nu)\mathbf{d}x^\mu\mathbf{d}x^\nu\\
&=e^\nu(1+H_0Y_{lm})\mathbf{d}t^2+2(H_1-r^{-2}e^{\lambda/2}\dot{W})\mathbf{d}t\mathbf{d}r\\
&\quad-r^2(1-KY_{lm})\mathbf{d}\Omega^2\\
&\quad-2r^{-2}e^{\lambda/2}W(Y_{lm,\theta}\mathbf{d}\theta+Y_{lm,\theta}\mathbf{d}\phi)\mathbf{d}r\end{aligned} \tag{59}$$

evaluated at the perturbed surface $r_i+\xi^r$. Hence, continuity of the intrinsic geometry adds four new boundary conditions:

$$0=[W]_{r_i}, \tag{60a}$$
$$0=[H_0]_{r_i}, \tag{60b}$$
$$0=[H_1]_{r_i}, \tag{60c}$$
$$0=[K]_{r_i}. \tag{60d}$$

Combining equations (60b) and (41) yields another useful result:

$$32\pi[\mu V]_{r_i}=[H_2]_{r_i}. \tag{61}$$

The boundary conditions provided by continuity of the extrinsic curvature turn out not to be useful to us; nevertheless, we give them here for completeness. The extrinsic curvature is expressed in terms of the 3-metric ${}^{(3)}\mathbf{g}$ by

$$\mathbf{K}=-\tfrac{1}{2}\pounds_{\mathbf{N}}{}^{(3)}\mathbf{g}, \tag{62}$$

where $\pounds_{\mathbf{N}}$ denotes the Lie derivative along $\mathbf{N}$:

$$\mathbf{K}_{\alpha\beta}=-\tfrac{1}{2}[{}^{(3)}\mathbf{g}_{\alpha\beta,\gamma}\mathbf{N}^\gamma+{}^{(3)}\mathbf{g}_{\mu\beta}\mathbf{N}^\mu_{\ ,\alpha}+{}^{(3)}\mathbf{g}_{\alpha\mu}\mathbf{N}^\mu_{\ ,\beta}]. \tag{63}$$

Continuity of the extrinsic curvature at the perturbed surface leads to two new boundary conditions:

$$0=[\tfrac{1}{2}\nu'H_2+H_0'+r^{-2}(\nu'e^{-\lambda/2})'W]_{r_i}, \tag{64a}$$
$$0=[H_2-rK'+2(r^{-1}e^{-\lambda/2})'W]_{r_i}. \tag{64b}$$

Finally, we have boundary conditions on the displacement itself. These depend on the nature of the media on either side of the interface. If the interface is a discontinuity in μ in an otherwise continuous elastic medium, then both the tangential and the radial displacement are continuous. Since the radial displacement is already known to be continuous (*cf.* equation 60a), this adds one new boundary condition:

$$[V]_{r_i}=0. \tag{65}$$

On the other hand, when the interface is between an elastic medium and a fluid then there is no constraint on the tangential fluid displacement and no additional boundary condition.

Are these boundary conditions sufficient to determine the displacement on one side of an interface given its value on the other? There are two types of interfaces, and we consider each separately.

At an elastic–elastic interface there are six independent boundary conditions: (60a–d), (65) and (58b). On either side of the interface, we have a sixth-order set of equations describing the pulsations; hence, the displacement on one side of the boundary completely determines the displacement on the other side.

The situation is more complicated at an elastic–fluid interface. First, consider the simplest case where we know the displacement in the elastic solid at the boundary. Then the four boundary conditions (60a–d) are sufficient to determine the displacement across the boundary and into the fluid (recall that W, H_0, H_1 and K determine V and H_2 in the fluid through equations 42 and 41). The conditions (58a) and (58b) are now conditions satisfied by the displacement in the elastic medium.

Now consider the case where we know the displacement in the fluid at the boundary with an elastic solid. Then equations (60a–d) provide four conditions, and (58b) provides a further condition on $(V/r^2)'$ in the elastic solid. The remaining boundary conditions do not further constrain the possible solutions on the elastic side of the interface: the discontinuity in V remains arbitrary and correspondingly there are two linearly independent solutions that satisfy the boundary conditions at the interface. At the distant boundary – on the other side of the solid for an elastic shell, or at the origin for an elastic core – however, there is only one combination of these two solutions that satisfies the no-stress boundary condition (58b). Thus, in order to determine the unique displacement on the solid side of the interface that matches to the displacement on the fluid side, it is necessary to integrate the perturbation equations throughout the solid. Nevertheless, the boundary conditions we have posed are sufficient to do this.

3.4.3 *The surface of the star*

The surface of the star is just another kind of interface, so that the boundary conditions given in Section 3.4.2 (specialized to the surface) apply here as well. The appropriate boundary conditions at $r=\mathscr{R}$ are

$$0=[H_0], \tag{66a}$$
$$0=[H_1], \tag{66b}$$
$$0=[K], \tag{66c}$$
$$0=\lim_{\varepsilon\to0+}H_0(\mathscr{R}+\varepsilon)-H_2(\mathscr{R}+\varepsilon), \tag{66d}$$
$$0=\left(\beta+\frac{e^{-\lambda/2}}{r^2}P'W-2\mu\mathscr{A}\right)_{\mathscr{R}-}, \tag{66e}$$
$$0=(\mu\mathscr{B})|_{\mathscr{R}-}. \tag{66f}$$

3.4.4 *Boundary conditions at infinity*

At large radius, we insist that the gravitational perturbation resolve itself into purely outgoing gravitational radiation (any incoming radiation would drive pulsations of the star). Since the EFE outside the star are independent of the stresses in the star, the boundary conditions are the same as those posed in Thorne & Campolattaro (1967a, section IIIc).

3.5 The eigenvalue problem

In order to see that these boundary conditions determine an eigenvalue problem, consider how we might go about constructing a solution to the differential equations. At the origin, regularity permits us two solutions. If we treat the perturbation equations as an initial value problem, then we may integrate each of these two solutions outward, passing through one or more interfaces where the shear modulus changes discontinuously, until the surface of the star is reached. At the surface, there will be a single linear combination of these two solutions that satisfies the boundary conditions on the Lagrangian pressure perturbation (*cf.* equation 66e) and the tangential stress (*cf.* equation 66f). The gravitational perturbations of this single solution may then be integrated to a large radius and resolved into ingoing and outgoing gravitational radiation. Only for certain choices of complex ω will the ingoing radiation vanish, and these are the eigenvalues of our system.

In actual practice, one technique of finding the eigenvalues of this system is not too different from the above description. This technique is commonly referred to as a 'shooting' (e.g. Bardeen, Thorne & Meltzer 1966), and its use in problems of relativistic stellar pulsations is common (Thorne 1969a; Finn 1986, 1987). Owing to numerical details it is inappropriate to integrate the system of equations directly from core to surface to infinity or through discontinuities, and this complicates the procedure. Similar problems arise in the study of stellar pulsations in the presence of density discontinuities. Finn (1987) shows how shooting may still be used to determine the eigenvalue of that system, and the techniques described there should also be of use here.

4 LAGRANGIAN DENSITY FOR NON-RADIAL PULSATIONS

4.1 Introduction

Since the EFE are derivable from a Lagrangian density, so are the perturbation equations for the non-radial pulsations. The Lagrangian and its associated variational principle provide another method for finding the eigenvalues and eigenmodes of the system, and are also useful for investigating the stability of the quasi-normal modes. We could obtain the Lagrangian directly from a second variation of the Hilbert action (Taub 1969; Moncrief 1974); instead, however, we shall reconstruct the Lagrangian from the field equations using the techniques described in Schumaker & Thorne (1983, appendix B). Intuitively, the steps required to construct the Lagrangian are simple to understand in terms of the Hilbert variational principle.

The Hilbert action is defined to be

$$I=\int_\Omega d^4x\left[\frac{1}{16\pi}(-g)^{1/2}R+\mathscr{L}_{\text{matter}}\right], \tag{67}$$

where R is the scalar curvature of space-time, $\mathscr{L}_{\text{matter}}$ is the Lagrangian density describing matter fields, g is the determinant of the fully covariant metric,

$$d^4x=dx^0\,dx^1\,dx^2\,dx^3 \tag{68}$$

is the product of coordinate differentials (and *not* the proper volume element), and Ω is a compact volume of space-time. The term $\mathscr{L}_{\text{geom}}=(-g)^{1/2}R/16\pi$ is sometimes associated with the geometry of space-time:

$$I_{\text{geom}}=\int_\Omega d^4x\,\mathscr{L}_{\text{geom}}=\frac{1}{16\pi}\int_\Omega d^4x(-g)^{1/2}R. \tag{69}$$

When the action is stationary with respect to metric variations $\delta\mathbf{g}_{\alpha\beta}$ that vanish at the boundary $\partial\Omega$, then $\mathbf{g}$ is a solution to the EFE. In particular,

$$\begin{aligned}\delta I&\equiv I(\mathbf{g}_{\alpha\beta}+\delta\mathbf{g}_{\alpha\beta})-I(\mathbf{g}_{\alpha\beta})\\&=-\frac{1}{16\pi}\int_\Omega d^4x(-g)^{1/2}(\mathbf{G}^{\alpha\beta}-8\pi\mathbf{T}^{\alpha\beta})\,\delta\mathbf{g}_{\alpha\beta}.\end{aligned} \tag{70}$$

In order to obtain this from the variation of the action (equation 67), we must integrate by parts. Since the $\delta\mathbf{g}_{\alpha\beta}$ vanishes on the boundary, the surface integrals vanish.

The formalism we have developed for non-radial pulsations is specialized to the Regge–Wheeler gauge, and we want to express the action in this gauge as well. Consequently, we cannot vary the metric components independently: it is only meaningful to consider variations $\delta\mathbf{g}_{\alpha\beta}$ expressible in Regge–Wheeler gauge. Since a general metric perturbation does not respect any specific gauge condition, we cannot recover all of the field equations (equations 22–28) by considering only variations of I with respect to Regge–Wheeler gauge metric variations; conversely we cannot reconstruct I from the EFE alone if we consider just Regge–Wheeler gauge metric variations. We need additional information related to our choice of gauge to reconstruct the action.

The information we seek is in the equations of motion. This is evident when we consider the special class of metric variations corresponding to coordinate gauge transformations. These are described by a displacement vector $\vec{\zeta}$ representing a coordinate change:

$$x^\mu_{\text{new}}=x^\mu_{\text{old}}+\zeta^\mu \tag{71a}$$

$$\begin{aligned}\delta\mathbf{g}_{\alpha\beta}&=\mathbf{g}^{\text{new}}_{\alpha\beta}-\mathbf{g}^{\text{old}}_{\alpha\beta}\\&=-(\zeta_{\alpha;\beta}+\zeta_{\beta;\alpha}).\end{aligned} \tag{71b}$$

The action I_{geom} is unchanged by these metric perturbations:

$$\begin{aligned}\delta I_{\text{geom}}&=\frac{1}{16\pi}\int_\Omega d^4x(-g)^{1/2}\mathbf{G}^{\alpha\beta}(\zeta_{\alpha;\beta}+\zeta_{\beta;\alpha})\\&=-\frac{1}{8\pi}\int_\Omega d^4x(-g)^{1/2}\mathbf{G}^{\alpha\beta}_{;\beta}\zeta_\alpha\end{aligned} \tag{72}$$

vanishes by virtue of the contracted Bianchi identities ($\mathbf{G}^{\alpha\beta}_{;\beta}=0$). Consequently, the variation of I (equation 67) with respect to the gauge transformation generated by $\vec{\zeta}$ yields the equations of motion:

$$\delta I=\int_{\Omega} d^4x(-g)^{1/2}\mathbf{T}^{\alpha\beta}_{;\beta}\zeta_{\alpha}. \tag{73}$$

An arbitrary metric variation $\delta\tilde{\mathbf{g}}_{\alpha\beta}$ can be considered a sum of a variation $\delta\mathbf{g}_{\alpha\beta}$ that satisfies our gauge condition and a gauge transformation, generated by $\vec{\zeta}$:

$$\delta\tilde{\mathbf{g}}_{\alpha\beta}=\delta\mathbf{g}_{\alpha\beta}-(\zeta_{\alpha;\beta}+\zeta_{\beta;\alpha}). \tag{74}$$

Thus, the variation of the action with respect to an arbitrary metric perturbation may be considered as a variation with respect to a gauge metric perturbation and a gauge transformation:

$$\begin{aligned}\delta I&=-\frac{1}{16\pi}\int_{\Omega} d^4x(-g)^{1/2}\delta\tilde{\mathbf{g}}_{\alpha\beta}(\mathbf{G}^{\alpha\beta}-8\pi\mathbf{T}^{\alpha\beta})\\ &=-\frac{1}{16\pi}\int_{\Omega} d^4x(-g)^{1/2}[\delta\mathbf{g}_{\alpha\beta}(\mathbf{G}^{\alpha\beta}-8\pi\mathbf{T}^{\alpha\beta})\\ &\quad-16\pi\zeta^{\alpha}\mathbf{T}^{;\beta}_{\alpha\beta}].\end{aligned} \tag{75}$$

The equations of motion thus provide the information needed to reconstruct the Hilbert action in any desired gauge. It remains only to associate the metric perturbation $\delta\mathbf{g}_{\alpha\beta}$ and the generator of coordinate gauge transformations $\vec{\zeta}$ with the generalized coordinates of the pulsation problem.

The gauge specific metric variation $\delta\mathbf{g}_{\alpha\beta}$ is readily expressible in terms of the generalized coordinates H_0, $\dot{H}_1$, H_2 and K, of the perturbed metric. If we identify $\vec{\zeta}$ with the Eulerian fluid displacement $\vec{\xi}$, then variations with respect to $\vec{\xi}$ will lead to the equations of motion and conversely, the equations of motion can be used together with the field equations to reconstruct the action.

4.2 The Lagrangian density

We now proceed to find the Lagrangian density for non-radial perturbations of a relativistic star with an isotropic shear modulus from the field equations (equations 22–28) and equations of motion (equations 34 and 35). First, we construct the quantity

$$\begin{aligned}\mathscr{L}\equiv-\frac{1}{16\pi}\int d\theta\, d\phi(-g)^{1/2}[\delta\mathbf{g}_{\alpha\beta}(\delta\mathbf{G}^{\alpha\beta}-8\pi\delta\mathbf{T}^{\alpha\beta})\\ +16\pi\zeta^{\alpha}\delta\mathbf{T}^{\beta}_{\alpha;\beta}],\end{aligned} \tag{76}$$

where

$$\begin{aligned}\delta\mathbf{g}_{\alpha\beta}\mathbf{d}x^{\alpha}\,\mathbf{d}x^{\beta}=-[e^{\nu}H_0^{\dagger}\,\mathbf{d}t^2+2\dot{H}_1^{\dagger}\,\mathbf{d}t\,\mathbf{d}r+e^{\lambda}H_2^{\dagger}\\ +r^2K^{\dagger}(\mathbf{d}\theta^2+\sin^2\theta\,\mathbf{d}\phi^2)]Y^{\dagger}_{lm}\end{aligned} \tag{77}$$

and

$$\zeta^r=\frac{\exp(-\lambda/2)}{r^2}W^{\dagger}Y^{\dagger}_{lm}, \tag{78}$$

$$\zeta_A=V^{\dagger}\frac{\partial Y^{\dagger}_{lm}}{\partial x^A}. \tag{79}$$

The quantities $H_0^{\dagger}$, $\dot{H}_1^{\dagger}$, $H_2^{\dagger}$, $K^{\dagger}$, $W^{\dagger}$ and $V^{\dagger}$ are arbitrary functions of r and t, and are (at this point) not assumed to be related to the generalized coordinates H_0, $\dot{H}_1$, H_2, K, W and V of the perturbation. They do, however, satisfy the boundary conditions described in Section 3.4. Now, we interpret $Y^{\dagger}_{lm}$ as the complex conjugate of Y_{lm} and evaluate the angular integrals in equation (76). The result is an expression for $\mathscr{L}$ which is a function of r and t only. Finally, we add to this expression a pure *coordinate* divergence so that the resulting expression is symmetric in 'daggered' and 'undaggered' quantities. [We add a coordinate divergence, as opposed to a covariant divergence, because of the way we have written the Lagrangian density: we removed the Jacobian $(-g)^{1/2}$ from the space-time volume element $(-g)^{1/2}d^4x$ and included it in our definition of the Lagrangian density; this was done precisely because it simplifies the mathematics by allowing us to work with coordinate divergences.] The resulting expression, with the distinction between daggered and undaggered quantities dropped, is the Lagrangian density:

$$\begin{aligned}\mathscr{L}&=r^2(\rho+P)e^{-(\nu-\lambda)/2}\left(\frac{W_{,t}}{r^2}\frac{W_{,t}}{r^2}+\frac{l(l+1)}{r^2}V_{,t}V_{,t}\right)\\ &\quad-2\mu r^2e^{(\nu+\lambda)/2}\int\Sigma^{\mu\nu}\Sigma_{\mu\nu}\sin\theta\, d\theta\, d\phi\\ &\quad-r^2e^{(\nu+\lambda)/2}\frac{\beta^2}{\gamma P}+\frac{e^{(\nu-\lambda)/2}}{r}\frac{P'}{\gamma P}SWW-2(\rho+P)e^{-\nu/2}\dot{H}_1W_{,t}\\ &\quad+\frac{e^{(\nu+\lambda)/2}}{16\pi}(l+2)(l-1)K(H_2-H_0)+\frac{r^2e^{(\nu-\lambda)/2}}{16\pi}(K'-2H_0')K'\\ &\quad-r^2e^{(\nu+\lambda)/2}H_0\alpha+\frac{e^{(\nu+\lambda)/2}}{16\pi}[8\pi r^2(\rho+P)-l(l+1)]H_0H_2\\ &\quad+\frac{re^{(\nu-\lambda)/2}}{8\pi}H_0'H_2-\frac{r^2e^{-(\nu-\lambda)/2}}{16\pi}[K_{,t}K_{,t}+2H_{2,t}K_{,t}]\\ &\quad+\frac{e^{-(\nu+\lambda)/2}}{16\pi}[4r\dot{H}_1H_{2,t}+l(l+1)\dot{H}_1\dot{H}_1]\\ &\quad+\frac{re^{(\nu-\lambda)/2}}{8\pi}\left[\left(1-\frac{r\nu'}{2}\right)H_0K'-\left(1+\frac{r\nu'}{2}\right)H_2K'\right]\\ &\quad+\frac{re^{-\nu/2}}{4\pi}(e^{-\lambda/2}r\dot{H}_1)'K_{,t}+\frac{e^{(\nu-\lambda)/2}}{16\pi}(1+r\nu')H_2H_2,\end{aligned} \tag{80}$$

where S is shorthand for the relativistic Schwarzschild convection discriminant,

$$S=P'-\gamma P\rho'/(\rho+P). \tag{81}$$

The generalized coordinates of the Lagrangian are H_0, $\dot{H}_1$, H_2, K, W and V: α is shorthand for the Eulerian density perturbation (equation 32), β is shorthand for the Eulerian pressure perturbation (equation 29), and $\mathbf{\Sigma}$ is shorthand for the stress tensor (equation 18a–d), all in terms of these generalized coordinates. When attempting to obtain the EFE or the equations of motion from L, or when using the variational principle to find the eigenmodes of a neutron star, it is important that α, β and σ be expanded in terms of the generalized coordinates H_0, $\dot{H}_1$, H_2, K, W and V before the variation is made.

In terms of L, the eigenmodes of a star with an isotropic shear modulus satisfy the variational equation

$$\delta\int_{\Omega} dt\, dr \mathscr{L} = 0, \tag{82}$$

where the integration is over a compact region of spacetime Ω and the perturbation is held fixed on the boundary $\partial\Omega$. The action $I \equiv \int_{\Omega} dt\, dr \mathscr{L}$ is perfectly suited for use in a Rayleigh–Ritz variational calculation for the quasi-normal eigenmodes of the system.

5 CONCLUSIONS

The presence of a solid crust or core in a neutron star leads to a set of elastic modes that are coupled to the star's more familiar fluid and gravitational modes. These modes correspond to the ability of the elastic solid crust or core to support shear waves. The non-radial elastic modes should couple most strongly to the star's *g*-modes, for which the fluid motion has a strong shear component. In addition to modifying the structure of the *g*-modes, however, the elastic solid composition of the neutron star leads to truly new modes which need not be strongly coupled to the fluid or gravitational modes of the star.

Preliminary investigations of the spectrum of non-radial modes in a neutron star with an elastic solid crust or core have proceeded in the absence of a fully relativistic pulsational theory by using a Cowling approximation that ignores the gravitational perturbations of the star and approximates the shear strain and stress. Here we present a fully relativistic theory of non-radial stellar pulsations that encompasses stars with an elastic crust or core. We have determined the shear strain of the solid constituent of the star in terms of the fluid and gravitational perturbation, and related it to the stress through an isotropic shear modulus (a bulk modulus could be included separately, or as an additional contribution to the isotropic pressure in an equation of state). This stress is included as a source to the Einstein Field Equations (EFE), above and beyond that owing to the fluid component of the star. We give the full set of EFE and equations of motion for the non-radial perturbation, and show that these reduce to a sixth-order set of ordinary differential equations for the radial dependence of the perturbation. Boundary conditions are posed at the star's centre and surface, and also at any places where the shear modulus changes discontinuously such as might occur within a single elastic solid or at interfaces between fluid and elastic solid. The boundary conditions, together with an outgoing gravitational wave boundary condition at infinity, determine an eigenvalue/eigenmode problem for the damped and undriven quasi-normal pulsational modes of the star, and we have described one method for solving this eigenvalue problem.

We have also constructed a Lagrangian density whose Euler–Lagrange equations are the field equations and equations of motion for the non-radial pulsations. This Lagrangian density is suited for use in a Rayleigh–Ritz solution of the eigenvalue problem for the quasi-normal pulsational modes, and may prove useful in a future investigation of the stability of the quasi-normal pulsational modes of the system.

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